Continuum Mechanics HW3

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Technion

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1 Starting Planar Poiseuille Flow

1.1 NS equation

In general the NS equation for an incompressible fluid is

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \boldsymbol{\nabla})\,\mathbf{v} = -\frac{1}{\rho}\boldsymbol{\nabla} p + \nu \nabla^2 \mathbf{v} \tag{1}$$

but the symmetries of the problem dictate that the velocity must be in the x direction and only depend on y and the time, and the given pressure gradient give

$$\frac{\partial v_x(y,t)}{\partial t} = \frac{\Delta P}{L\rho} + \nu \frac{\partial^2 v_x(y,t)}{\partial y^2} \tag{2}$$

The boundary conditions are

$$v_x(y=0,t) = v_x(y=h,t) = 0 \tag{3}$$

and the initial conditions are

$$v_x(y,t=0) = 0 \tag{4}$$

1.2 Steady State Solution

The steady state solution does not depend on time therefore we find it from

$$\frac{\Delta P}{L\rho} + \nu \frac{\partial^2 v_x(y,t)}{\partial y^2} = 0 \implies v_x(y,t) = -\frac{\Delta P}{2L\rho\nu} y^2 + Ay + B \tag{5}$$

I noticed I missed the factor of 2 here then I added it back manually to a bunch of places I think it should have been. If there are mistakes of a missing or superfluous 2 it's because of that I'm sorry :(
Using the boundary conditions we have

$$B = 0, \quad A = \frac{h\Delta P}{2L\rho\nu} \tag{6}$$

so

$$v_x^0 = -\frac{\Delta P}{2L\rho\nu} y^2 + \frac{2h\Delta P}{L\rho\nu} y = \frac{\Delta P}{2L\rho\nu}\left(hy - y^2\right) \tag{7}$$

1.3 New Variable

noticing that

$$\frac{\partial v_x(y,t)}{\partial t} = \frac{\Delta P}{L\rho} + \nu \frac{\partial^2 v_x(y,t)}{\partial y^2} \tag{8}$$

and

$$\frac{\partial v_x^0(y)}{\partial t} = \frac{\Delta P}{L\rho} + \nu \frac{\partial^2 v_x^0(y)}{\partial y^2} \tag{9}$$

can be rearranged to

$$\left(\frac{\partial}{\partial t} - \nu \frac{\partial^2}{\partial y^2}\right) v_x = \frac{\Delta P}{L\rho} \qquad \left(\frac{\partial}{\partial t} - \nu \frac{\partial^2}{\partial y^2}\right) v_x^0 = \frac{\Delta P}{L\rho} \tag{10}$$

and if we subtract the equations we get

$$\left(\frac{\partial}{\partial t} - \nu\frac{\partial^2}{\partial y^2}\right) v'_x = 0 \tag{11}$$

which can be rearranged to

$$\frac{\partial v'_x}{\partial t} = \nu\frac{\partial^2 v'_x}{\partial y^2} \tag{12}$$

which is the diffusion equation. The initial conditions are

$$v'_x(t=0, y) = -v^0_x \tag{13}$$

1.4 Separation of Variables

by replacing

$$v'_x = \Theta(t)Y(y) \tag{14}$$

we have

$$\Theta' Y = \nu\Theta Y'' \implies \frac{\Theta'}{\Theta} = \nu\frac{Y''}{Y} \tag{15}$$

meaning both sides are constants, so

$$\frac{Y''}{Y} = -\lambda^2 \implies Y = Ae^{i\lambda y} + Be^{-i\lambda y} \tag{16}$$

using the boundary conditions we have immediately

$$A = -B \tag{17}$$

thus

$$2A\sinh(\lambda h) = 0 \implies \lambda = \frac{\pi n}{h} \tag{18}$$

giving us

$$Y = A\sin\left(\frac{\pi n y}{h}\right) \tag{19}$$

and we get the value of A from normalization of the function set

$$\|Y\| = \int_0^h A^2\sin^2\left(\frac{\pi n y}{h}\right)\mathrm{d}y = \frac{A^2 h}{2} = 1 \implies A = \sqrt{\frac{2}{h}} \tag{20}$$

so

$$Y = \sqrt{\frac{2}{h}}\sin\left(\frac{\pi n y}{h}\right) \implies Y'' = -\sqrt{\frac{2}{h}}\frac{\pi^2 n^2}{h^2}\sin\left(\frac{\pi n y}{h}\right) \tag{21}$$

we can now plug this back into the equation

$$\Theta' = -\nu\frac{\pi^2 n^2}{h^2}\Theta \tag{22}$$

so

$$\Theta = Ce^{-\nu\left(\frac{\pi n}{h}\right)^2 t} \tag{23}$$

plugging this back into the initial conditions

$$\sum_{n=1}^{\infty}\sqrt{\frac{2}{h}}C_n\sin\left(\frac{\pi n y}{h}\right)e^{-\nu\left(\frac{\pi n}{h}\right)^2 t}=\frac{\Delta P}{2L\rho\nu}\left(y-h\right)y \tag{24}$$

but since it's the initial condition then $t=0$ so

$$\sum_{n=1}^{\infty}C_n\sin\left(\frac{\pi n y}{h}\right)=\sqrt{\frac{h}{2}}\frac{\Delta P}{2L\rho\nu}\left(y-h\right)y \tag{25}$$

We find C_n by

$$C_n=\sqrt{\frac{2}{h}}\int_0^h \mathrm{d}y\sin\left(\frac{\pi n y}{h}\right)\frac{\Delta P}{L\rho\nu}\left(y-h\right)y \tag{26}$$

$$=\sqrt{\frac{2}{h}}\frac{\Delta P}{L\rho\nu}\int_0^h \mathrm{d}y\sin\left(\frac{\pi n y}{h}\right)\left(y-h\right)y \tag{27}$$

throwing this into an inegral solver we get

$$C_n=\begin{cases}0 & n \text{ even}\\ \sqrt{\frac{2}{h}}\frac{4h^2\Delta P}{\rho\nu L\pi^3 n^3} & n \text{ odd}\end{cases} \tag{28}$$

so

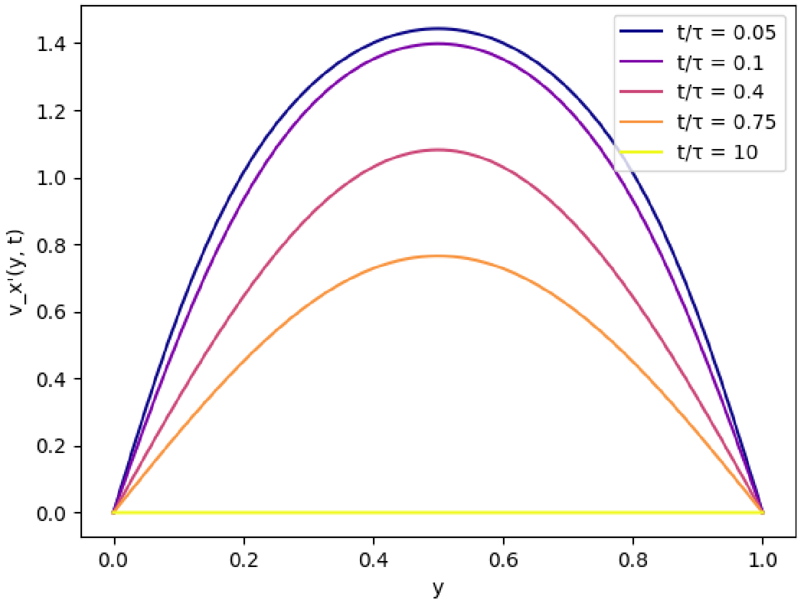
$$v_x'(y,t)=\sum_{n=0}^{\infty}\sqrt{\frac{2}{h}}\frac{4h^2\Delta P}{2\rho\nu L\pi^3\left(2n+1\right)^3}\sin\left(\frac{\pi\left(2n+1\right)y}{h}\right)e^{-\nu\left(\frac{\pi(2n+1)}{h}\right)^2 t} \tag{29}$$

1.5 Time Scale and Plot

Looking at the exponent of the solution we can see the timescale is

$$\tau = \frac{1}{\nu}\left(\frac{h}{\pi\,(2n+1)}\right)^2 \tag{30}$$

Which in a plot looks like



2 Momentum and Energy Fluxes in Poiseuille Flow

2.1 Nonvanishing terms

In the lectures we saw that for an incompressible Newtonian fluid

$$\Pi_{ij} = \rho v_i v_j + p\delta_{ij} - 2\mu\dot{\varepsilon}_{ij} = \rho v_i v_j + p\delta_{ij} - \mu\left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i}\right) \tag{31}$$

so in matrix form

$$\Pi_{ij} = \begin{pmatrix} \rho v_x^2 + p - 2\mu\partial_x v_x & \rho v_x v_y - \mu\left(\partial_y v_x + \partial_x v_y\right) \\ \rho v_x v_y - \mu\left(\partial_y v_x + \partial_x v_y\right) & \rho v_y^2 + p - 2\mu\partial_y v_y \end{pmatrix} \tag{32}$$

I'm going to assume that this question refers to the same kind of geometry as the previous question, which lets me simplify and substitute

$$\Pi_{ij} = \begin{pmatrix} \rho v_x^2 + p & -\mu\partial_y v_x \\ -\mu\partial_y v_x & p \end{pmatrix} = \begin{pmatrix} \rho\left(\frac{\Delta P}{2L\rho\nu}\left(hy - y^2\right)\right)^2 + p_0 - \frac{\Delta P}{L}x & -\frac{\Delta P}{2L}\left(h - 2y\right) \\ \frac{\Delta P}{2L}\left(h - 2y\right) & p_0 - \frac{\Delta P}{L}x \end{pmatrix} \tag{33}$$

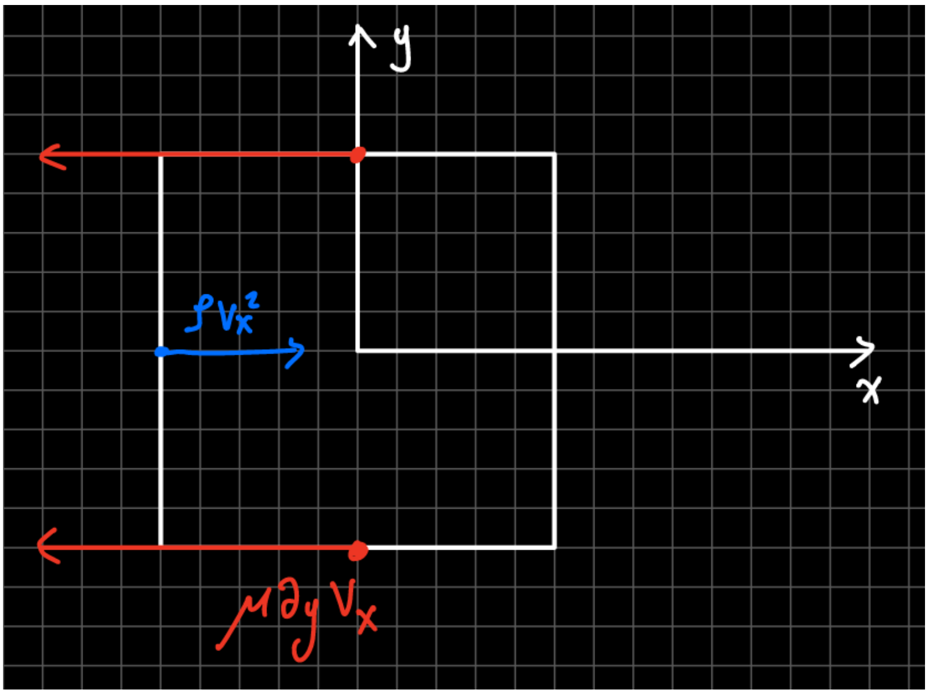
2.2 Change in time and balance

first

$$\partial_x \Pi_{xx} = -\frac{\Delta P}{L} \qquad\qquad \partial_y \Pi_{xy} = -2 \cdot -\frac{\Delta P}{2L} = \frac{\Delta P}{L} \tag{34}$$

So they are of equal magnitude and opposite sign. also

$$\partial_t\left(\rho v_x\right) = \partial_x \Pi_{xx} + \partial_y \Pi_{xy} = 0 \tag{35}$$



2.3 Heat

The internal energy is

$$\partial_t (\rho E) + \partial_{x_k} (v_k \rho E - \kappa \partial_{x_k} T) = \sigma_{ik} \frac{\partial v_i}{\partial x_k} \tag{36}$$

Since we have Poiseuille flow we can eliminate the elements depending on the derivative of the velocity since in both the case of x and y its zero

$$\partial_t (\rho E) - \kappa \left(\partial_x^2 + \partial_y^2\right) T = \sigma_{xy} \frac{\partial v_x}{\partial y} \tag{37}$$

assuming the energy is conserved and using the symmetry for the x axis

$$-\kappa \partial_y^2 T = \sigma_{ik} \frac{\partial v_x}{\partial y} \tag{38}$$

or

$$\kappa \partial_y^2 T + \mu \left(\frac{\partial v_x}{\partial y}\right)^2 = 0 \tag{39}$$

The flow dissipates the energy but the energy is transported away, meaning at each place it remains constant over time.

2.4 Steady State profile

we can insert the velocity profile found earlier

$$\kappa \partial_y^2 T + \mu \left(\frac{\Delta P}{2L\rho\nu} (h - 2y)\right)^2 = 0 \implies \partial_y^2 T = -\frac{\mu}{\kappa} \left(\frac{\Delta P}{2L\rho\nu}\right)^2 \left(h^2 - 4hy + 4y^2\right) \tag{40}$$

therefore

$$T = -\frac{\mu}{\kappa} \left(\frac{\Delta P}{2L\rho\nu}\right)^2 \left(\frac{2}{3} hy^3 - \frac{1}{3} y^4 - \frac{1}{2} h^2 y^2 + Ay\right) + B \tag{41}$$

boundary conditions

$$T(h/2) = T_0 \qquad \partial_y T(y = h/2) = 0 \tag{42}$$

so

$$-\frac{\mu}{\kappa} \left(\frac{\Delta P}{2L\rho\nu}\right)^2 \left(2h (h/2)^2 - \frac{4}{3} (h/2)^3 - h^2 (h/2) + A\right) = 0 \implies A = \frac{h^3}{6} \tag{43}$$

and

$$-\frac{\mu}{\kappa} \left(\frac{\Delta P}{2L\rho\nu}\right)^2 \left(\frac{2}{3} h (h/2)^3 - \frac{1}{3} (h/2)^4 - \frac{1}{2} h^2 (h/2)^2 + \frac{h^3}{6} (h/2)\right) + B = T_0 \tag{44}$$

gives

$$B = T_0 - \frac{\mu}{\kappa} \left(\frac{\Delta P}{2L\rho\nu}\right)^2 \left(\frac{1}{48} h^2 \left(7h^2 - 6\right)\right) \tag{45}$$

thus

$$T = -\frac{\mu}{\kappa} \left(\frac{\Delta P}{2L\rho\nu}\right)^2 \left(\frac{2}{3} hy^3 - \frac{1}{3} y^4 - \frac{1}{2} h^2 y^2 + \frac{h^3}{6} y\right) + T_0 - \frac{\mu}{\kappa} \left(\frac{\Delta P}{2L\rho\nu}\right)^2 \left(\frac{1}{48} h^2 \left(7h^2 - 6\right)\right) \tag{46}$$

2.5 Estimate

I'm very short on time so I just plugged the numbers without typing everything out

$$\Delta T \approx 1.4 \times 10^{-2}\mathrm{K} \tag{47}$$

3 Flow around a sphere in small reynolds

for $r \gg R$

$$p = c_1 F \frac{\cos\theta}{r^2} \qquad v_r = c_2 \frac{F}{\mu}\frac{\cos\theta}{r} \qquad v_\theta = c_3 \frac{F}{\mu}\frac{\sin\theta}{r} \tag{48}$$

momentum conservation

$$\partial_t (\rho v_i) + \partial_{x_k} (\rho v_i v_k - \sigma_{ik}) = 0 \tag{49}$$

incompressibility applied to the velocity gives

$$\boldsymbol{\nabla} \cdot v = \frac{1}{r^2}\partial_r \left(r^2 v_r\right) + \frac{1}{r\sin\theta}\partial_\theta \left(\sin\theta v_\theta\right) = c_2 \frac{F}{\mu}\frac{\cos\theta}{r^2} + c_3 \frac{F}{\mu}\frac{2\cos\theta}{r^2} = 0 \implies c_2 = -2c_3 \tag{50}$$

and assuming the solution is steady over time we get

$$\partial_{x_k} (\rho v_i v_k - \sigma_{ik}) = 0 \tag{51}$$

looking at the element in the direction of θ

$$\partial_{x_k} (\rho v_\theta v_k - \sigma_{\theta k}) = \frac{1}{r^2}\partial_r \left(r^2 (\rho v_\theta v_r) - \sigma_{\theta r}\right) + \frac{1}{r\sin\theta}\partial_\theta \left(\sin\theta \left(\rho v_\theta^2 - \sigma_{\theta\theta}\right)\right) = 0 \tag{52}$$

but

$$\sigma_{\theta\theta} = -p + 2\mu \left(\frac{1}{r}\partial_\theta v_\theta + \frac{1}{r} v_r\right) \tag{53}$$

so

$$\frac{1}{r\sin\theta}\partial_\theta \left(\sin\theta \left(\rho \left(c_3 \frac{F}{\mu}\frac{\sin\theta}{r}\right)^2 + c_1 F \frac{\cos\theta}{r^2} - 2\mu \left(\frac{1}{r}\partial_\theta \left(c_3 \frac{F}{\mu}\frac{\sin\theta}{r}\right) + \frac{1}{r}\left(c_2 \frac{F}{\mu}\frac{\cos\theta}{r}\right)\right)\right)\right) = 0 \tag{54}$$

we can see every element has 2 sin or cos functions multiplied, together with the fact this can't depend on θ (and because I want to at least try to finish this homework set which is extremely long) I'll treat them all as just 1.

$$\rho \left(c_3 \frac{F}{\mu}\frac{1}{r}\right)^2 + c_1 F \frac{1}{r^2} - 2\mu \left(\frac{1}{r} c_3 \frac{F}{\mu}\frac{1}{r} + \frac{1}{r}\left(c_2 \frac{F}{\mu}\frac{1}{r}\right)\right) = 0 \tag{55}$$

simplifying

$$\rho c_3^2 \frac{F^2}{\mu^2}\frac{1}{r^2} + c_1 F \frac{1}{r^2} - 2c_3 F \frac{1}{r^2} - 2c_2 F \frac{1}{r^2} = 0 \tag{56}$$

then

$$\rho c_3^2 \frac{F}{\mu^2}\frac{1}{r^2} + c_1 \frac{1}{r^2} - 2c_3 \frac{1}{r^2} - 2c_2 \frac{1}{r^2} = 0 \tag{57}$$

the first element depends on sine squared so for some values of theta, incl. 0, it vanishes, so we can solve for

$$c_1 - 2c_3 - 2c_2 = 0 \tag{58}$$

giving us

$$c_1 = 2c_2 + 2c_3 = 2c_3 - 4c_3 = -2c_3 \tag{59}$$

so the relation is

$$c_1 = c_2 = -2c_3 \tag{60}$$

4 Torque by rotating sphere

4.1 Dimensional Analysis

R has dimension of length, Ω of 1 over time, and μ of mass over (length times time). Torque has dimensions of mass times length squared divided by time squared. therefore

$$\tau = A\mu\Omega R^3 \tag{61}$$

4.2 Symmetry

The only preferred direction in this problem is generated by the rotation of the sphere. If we move to a rotating frame of reference whose rotation is centered on the center of the sphere, it would appear as if it's in perfect spherical symmetry all around. Therefore, the only direction the velocity of the fluid around the sphere could have is the ϕ direction. But, even then we have symmetry for rotation around ϕ, meaning the velocity can't actually depend on ϕ. In total

$$\mathbf{v} = v(r,\theta)\hat{\phi} \tag{62}$$

4.3 Solution

Since we're in low reynolds, we'll use the stokes equation

$$\boldsymbol{\nabla} p = \mu\nabla^2\mathbf{v} \tag{63}$$

rewriting it as

$$\nabla^2 v_\phi - \frac{v_\phi}{r^2\sin^2\theta} = 0 \tag{64}$$

now applying Separation of Variables

$$v_\phi = R(r)\Theta(\theta) \tag{65}$$

so

$$\begin{aligned}
\nabla^2 v_\phi &= \frac{1}{r^2}\partial_r\left(r^2\partial_r\left(R\Theta\right)\right) + \frac{1}{r^2\sin\theta}\partial_\theta\left(\sin\theta\partial_\theta\left(R\Theta\right)\right) &(66)\\
&= \frac{1}{r^2}\partial_r\left(r^2R'\Theta\right) + \frac{1}{r^2\sin\theta}\partial_\theta\left(\sin\theta R\Theta'\right) &(67)\\
&= \frac{\Theta}{r^2}\left(2rR' + r^2R''\right) + \frac{R}{r^2\sin\theta}\left(\cos\theta\Theta' + \sin\theta\Theta''\right) &(68)
\end{aligned}$$

so

$$\frac{R\Theta}{r^2\sin^2\theta} = \frac{\Theta}{r^2}\left(2rR' + r^2R''\right) + \frac{R}{r^2\sin\theta}\left(\cos\theta\Theta' + \sin\theta\Theta''\right) \tag{69}$$

or

$$\frac{1}{r^2\sin^2\theta} = \frac{1}{Rr^2}\left(2rR' + r^2R''\right) + \frac{1}{\Theta r^2\sin\theta}\left(\cos\theta\Theta' + \sin\theta\Theta''\right) \tag{70}$$

or

$$\frac{1}{\sin^2\theta} = \frac{1}{R}\left(2rR' + r^2R''\right) + \frac{1}{\Theta\sin\theta}\left(\cos\theta\Theta' + \sin\theta\Theta''\right) \tag{71}$$

or

$$\frac{1}{\sin^2\theta} - \frac{\cos\theta\Theta' + \sin\theta\Theta''}{\Theta\sin\theta} = \frac{2rR' + r^2R''}{R} \tag{72}$$

or

$$-\frac{\sin\theta\cos\theta\Theta' + \sin^2\theta\Theta'' + \Theta}{\Theta\sin^2\theta} = \frac{2rR' + r^2R''}{R} = -\lambda^2 \tag{73}$$

now we can use the separation solution process

This homework set was simply too long and I could not finish everything by the time I needed to submit it. Everything would have likely taken me at least 20 pages and I have other courses too. I submit this knowing I'll get only partial marks, I have no choice.